So how does virtualization help here? Well, before actually answering the question, we need to look at the service models of a cloud. A cloud basically offers itself as one of the three:

- **Infrastructure as a service (IaaS):** This consists of the hardware and networking services such as computers, firewalls, routers etc.
- **Platform as a service (PaaS):** This consists of the infrastructure plus the platform. Here the platform refers to the operating system, programming language execution environment, software servers such as *FTP, HTTP,* email servers etc. A cloud offering a LAMP stack can be thought of as a PaaS cloud.
- **Software as a Service (SaaS):** This includes the infrastructure, platform and the software running on top. For example, a cloud with a blogging platform can be visualized as a SaaS cloud. Blogpost.com can be thought of as a SaaS because you don't have to care about what hardware, networking devices, programming languages or databases are in use behind the scene. You just have to use the software therein!

Virtualization can help in accomplishing the required actions easily for all the three models:

- Since all the hardware is virtual, you can have many virtual machines on many physical servers and that will form the backbone of a computational resource pool. Since all virtual machines can be managed with the virtualization software, maintenance and monitoring of resources becomes easier.
- Storage devices are virtual as well. Virtualization can help automate the creation of new disk drives on the fly and that can help expand everything! Also, all those storage devices can be managed with the virtualization software, hence making the "on-demand" and "resource pooling" characteristics of the cloud come near reality!
- Since networking between virtual machines is also handled by virtualization software, you can add and configure new network interfaces manually or automatically. Once again, a lot of ease! By points mentioned till now, we can say that virtualization can help IaaS model clouds!
- Installation of operating systems inside virtual machines can be automated. Also, with snapshot and cloning features, creation of new virtual machines can easily be accomplished. A pre-configured VM can be cloned and kept aside and can then be repeatedly used to create new virtual machines on the fly. On-demand, eh? Since all the components